

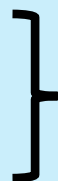
# Voice Disorder<sup>1</sup>



Objective  
Acoustic  
Parameter



Subjective  
Patient Report



Heterogeneity

Causes

Patients

Symptoms



Interpretable features  
bridge to clinical care

No MFCCs



Frequency, spectral, time,  
loudness-based  
Diagnosis and Treatment

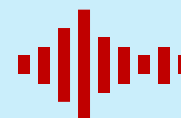


Screening ML

|                   |                   |
|-------------------|-------------------|
|                   | FALSE<br>POSITIVE |
| FALSE<br>NEGATIVE |                   |

Error analysis

FP:



CPP

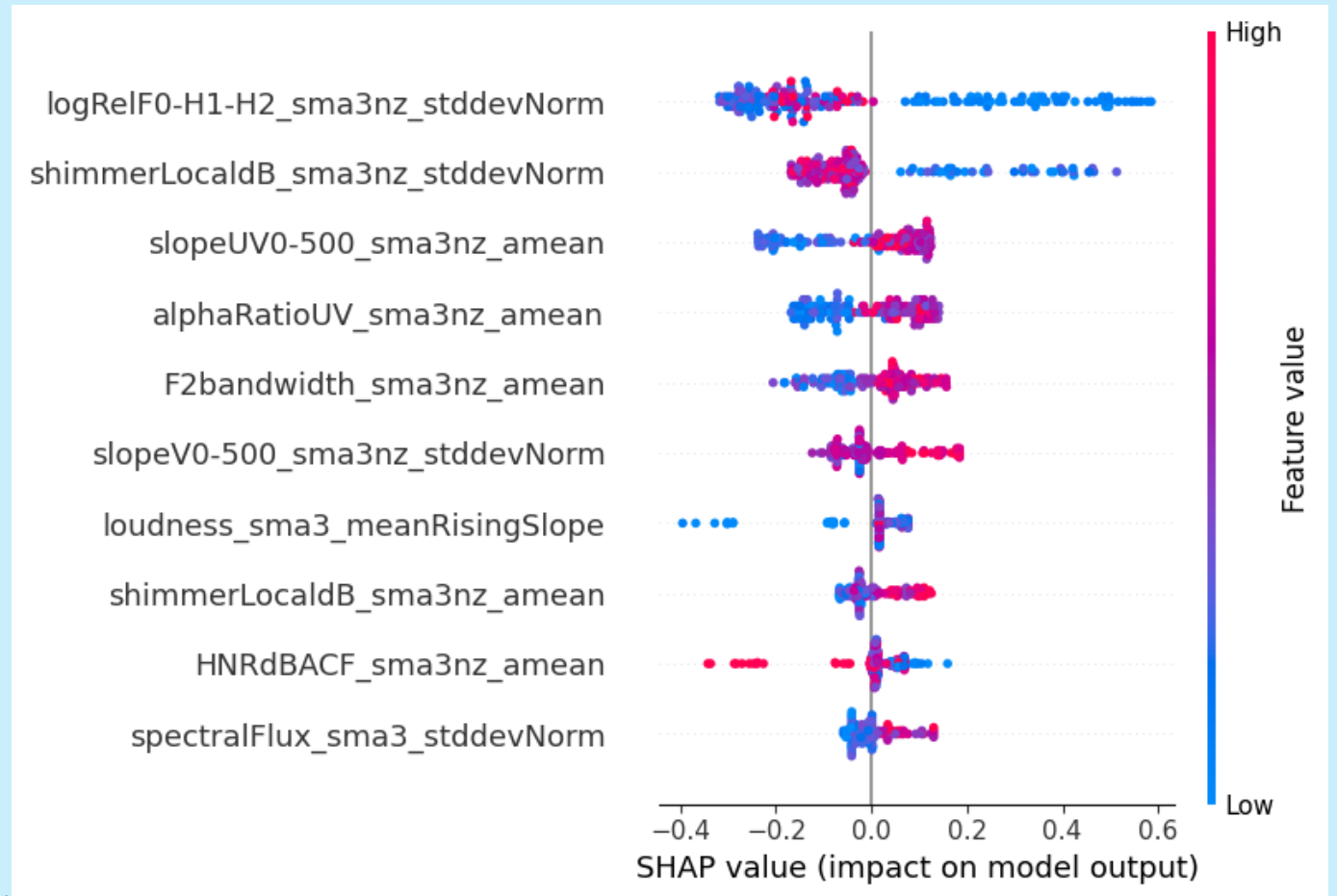
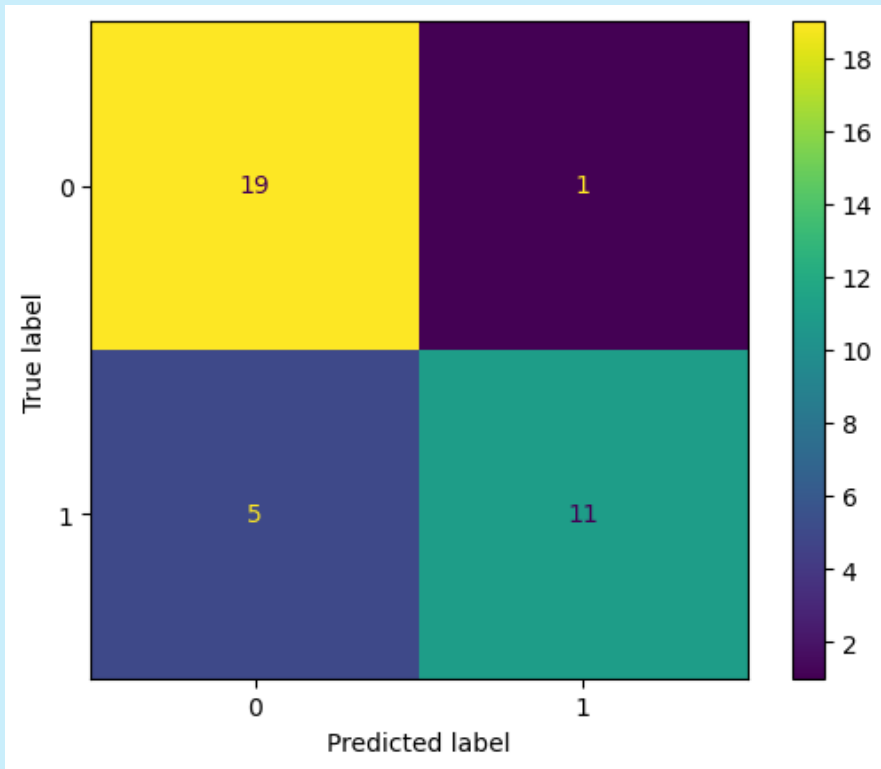
Cepstral peak  
prominence

FN:



VHI

Voice  
Handicap  
Index



XGBoost using 3-fold stratified cross validation

accuracy: 83.33%

precision: 0.854

recall: 0.819

F1: 0.786

**FP:**

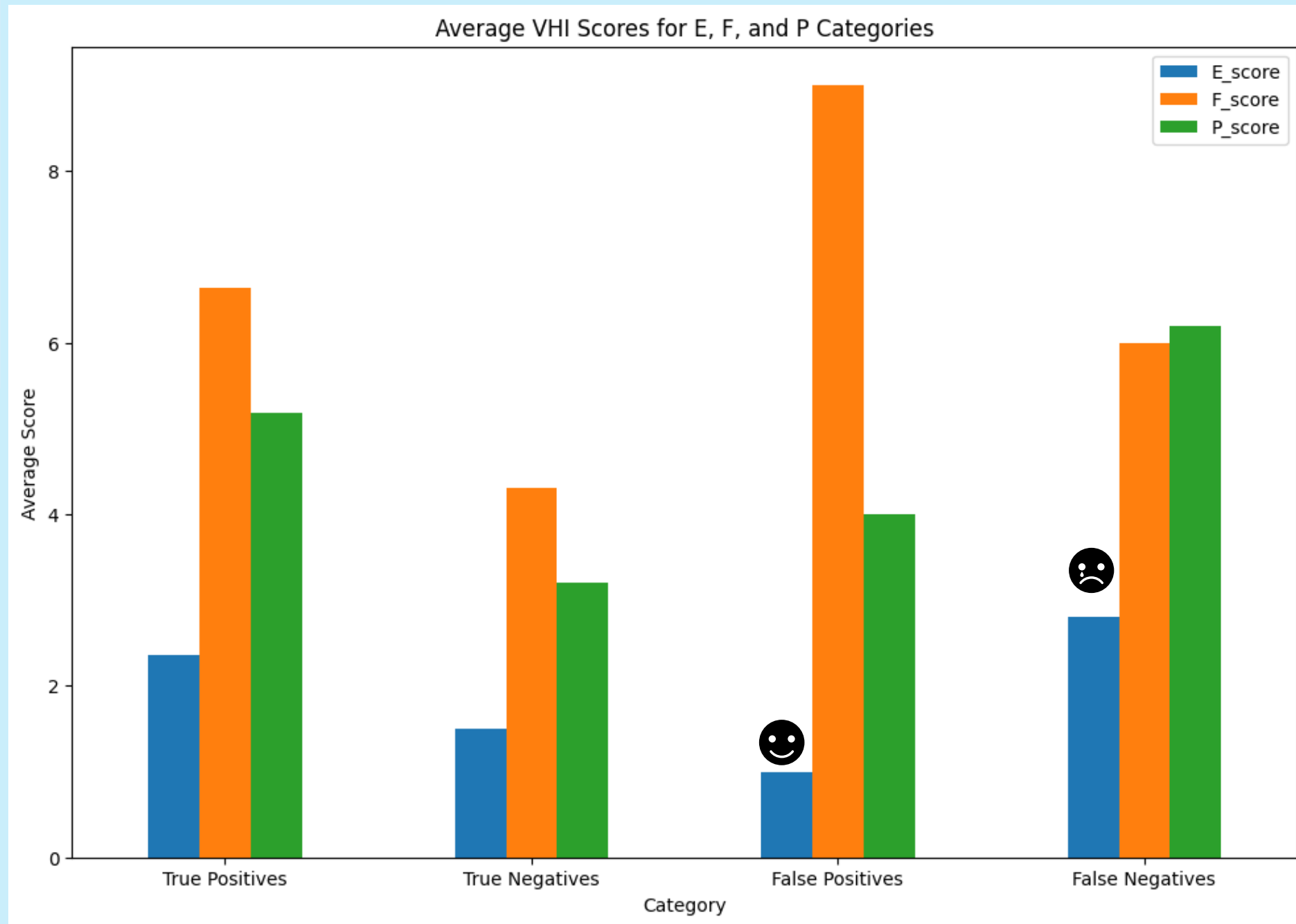
 Matched

 Unmatched

**FN:**

 Unmatched

 Matched



FP:



Matched



Unmatched

FN:



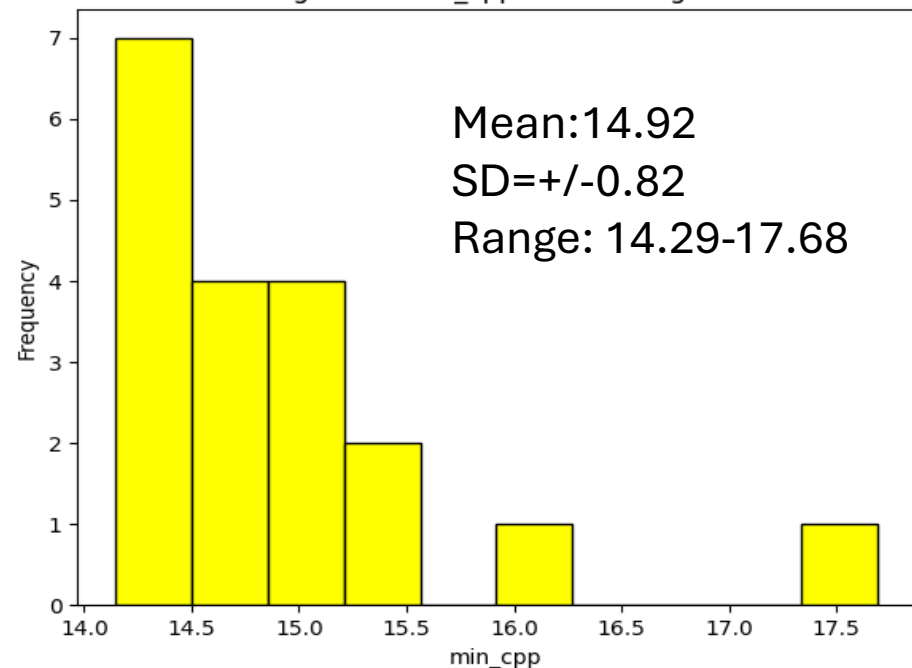
Unmatched



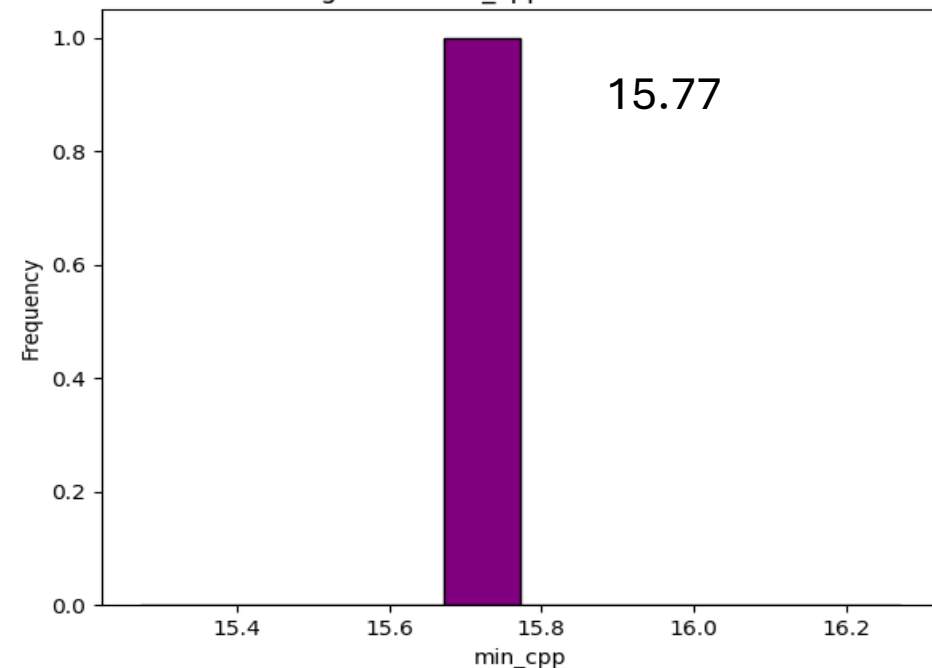
Matched

Mean:15.84  
SD= +/- 0.57  
Range: 15.17-16.76

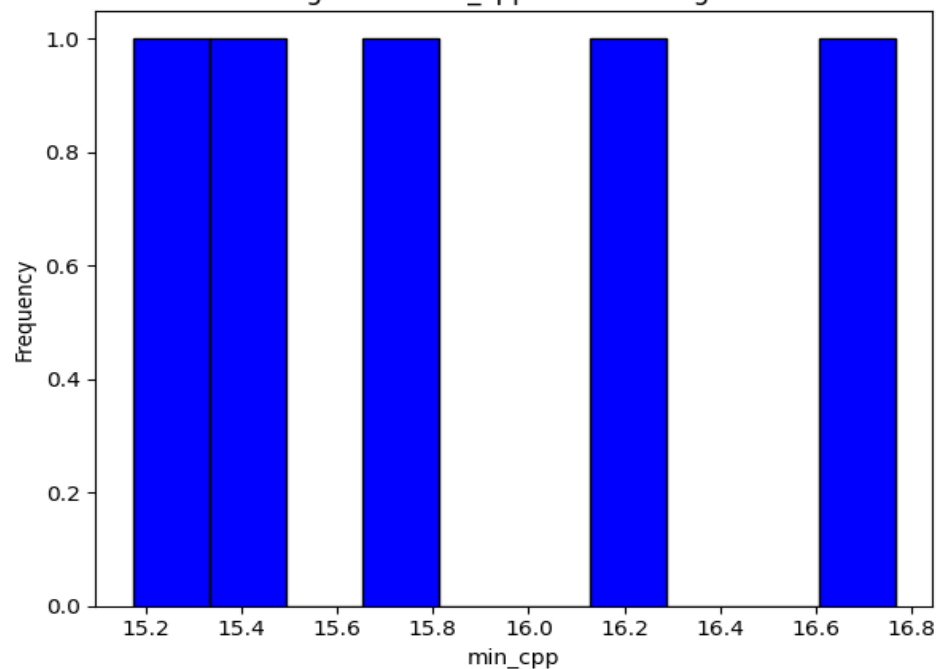
Histogram of min\_cpp for True Negatives



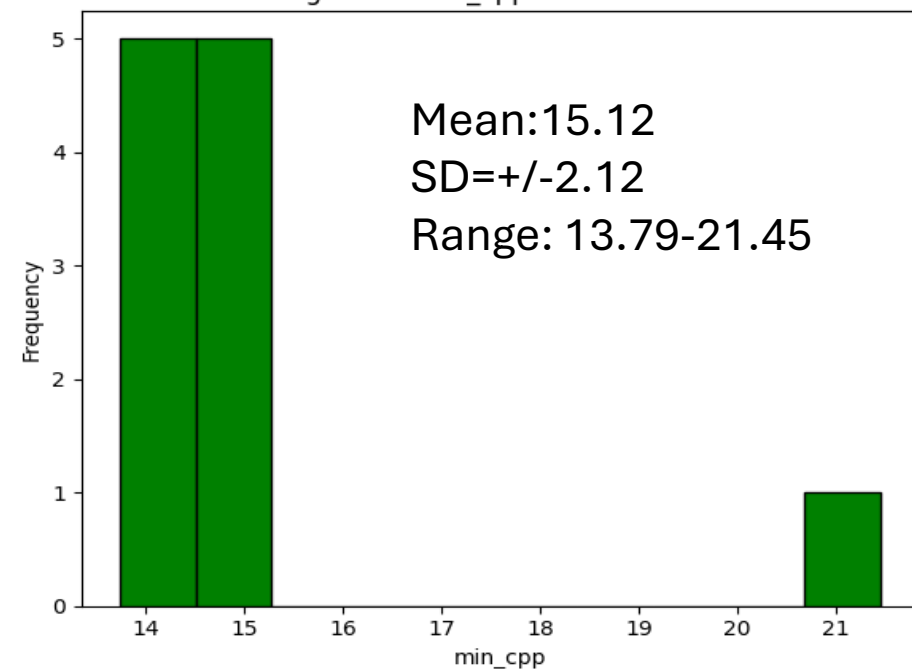
Histogram of min\_cpp for False Positives



Histogram of min\_cpp for False Negatives



Histogram of min\_cpp for True Positives





- Heterogeneity in acoustic (objective) voice quality



- True Controls?
- Mild but significant features?



- Putting VHI in the model:
  - Random Forest, SVM vs. XGBoost



- Patient Motivation to seek care

### **Future Directions:**

#### **•Validate and Confirm:**

- Our hypothesis of false positives (FP) and false negatives (FN)

#### **•Improve Data Quality:**

- Increase size of training and validation datasets
- Include diverse disorder types in datasets

#### **•Enhance Reliability:**

- Ensure reliability of acoustic parameters

#### **•Model Optimization:**

- Fine-tune our model
- Use other datasets for evaluation

|                                 |                                           | Missing | Overall     | False       | True        | P-Value |
|---------------------------------|-------------------------------------------|---------|-------------|-------------|-------------|---------|
| n                               |                                           |         | 176         | 97          | 79          |         |
| age, mean (SD)                  |                                           | 0       | 56.1 (18.3) | 53.2 (18.9) | 59.6 (17.0) | 0.019   |
| gender_identity, n (%)          | Female gender identity                    |         | 111 (63.1)  | 63 (64.9)   | 48 (60.8)   | 0.070   |
|                                 | Male gender identity                      |         | 60 (34.1)   | 29 (29.9)   | 31 (39.2)   |         |
|                                 | Non-binary or genderqueer gender identity |         | 5 (2.8)     | 5 (5.2)     |             |         |
| is_control_participant_x, n (%) | No                                        |         | 157 (89.2)  | 97 (100.0)  | 60 (75.9)   | <0.001  |
|                                 | Yes                                       |         | 19 (10.8)   |             | 19 (24.1)   |         |